

What are the latest things known about multimodal rag?

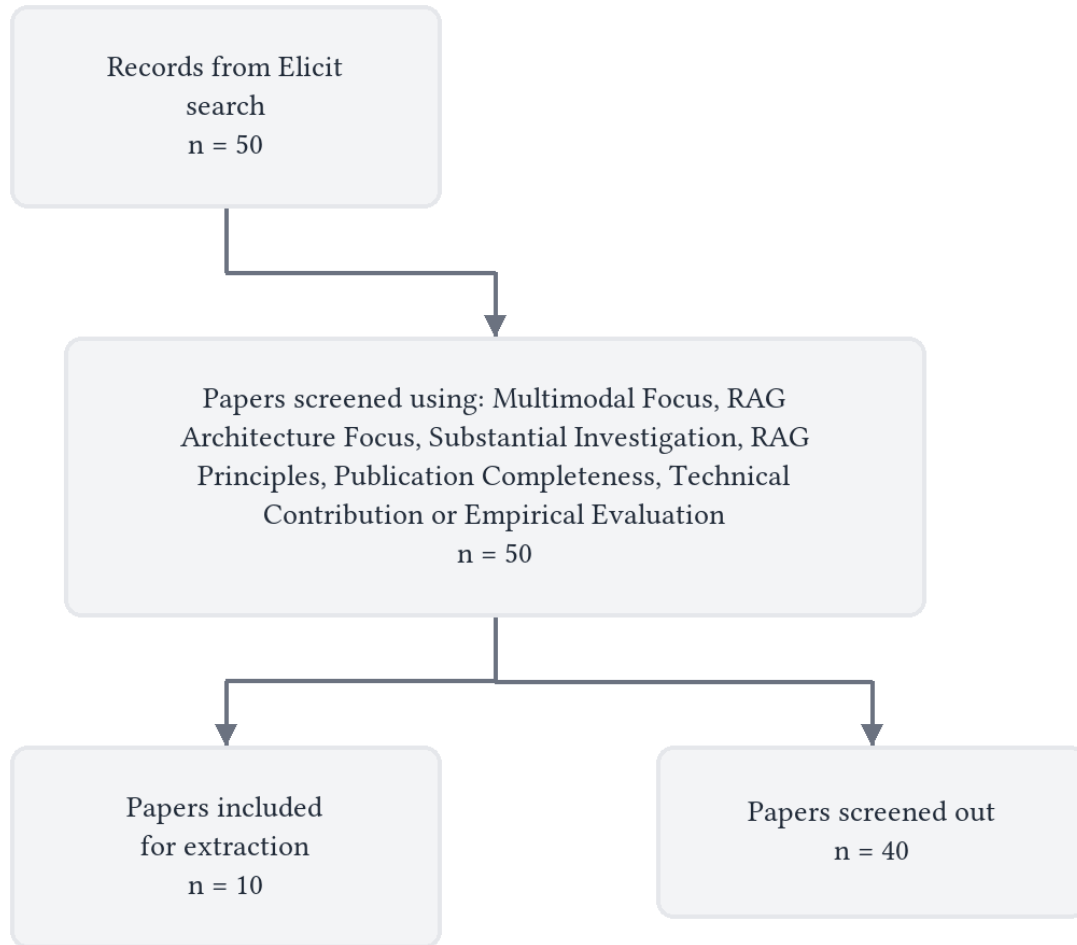
The latest advances in multimodal RAG demonstrate substantial performance improvements through hierarchical multi-agent and graph-based architectures, while recent findings reveal critical security vulnerabilities and the counterintuitive discovery that textual image summaries currently outperform direct multimodal embeddings despite failing on visual attribute questions.

Abstract

Recent multimodal RAG systems achieve substantial performance improvements over text-only approaches, with medical applications showing the largest gains (43.8% improvement in factual accuracy [1]) and document understanding systems demonstrating increasing advantages on longer documents (13+ point improvements for documents exceeding 100 pages [2]). Three architectural paradigms dominate current implementations: agent-based systems using ReAct logic for multi-hop reasoning [3], graph-based approaches integrating structured knowledge traversal [4], and hybrid designs combining both strategies [2, 5]. The highest-performing systems integrate textual, visual, and structured data through specialized retrieval mechanisms, achieving 92% accuracy with complete modality coverage [3] and state-of-the-art results in zero-shot settings [5]. However, counterintuitively, textual summaries of images currently outperform direct multimodal embeddings for integration [6], though this approach fails for visual attribute questions about colors or shapes [3].

Critical challenges remain unresolved, including severe security vulnerabilities where knowledge poisoning attacks achieve 56% success rates in manipulating outputs [7] and can completely nullify generation with a single injected entry [7]. Cross-modal alignment issues persist, with systems struggling to establish coherent correlations between visual and textual elements [5] while exhibiting modality biases toward text [8]. Computational efficiency remains problematic, particularly for video processing and long-context documents [8, 9]. Solutions under development include domain-aware retrieval mechanisms [1], consistency voting for multi-source integration [5], and dual-graph construction to capture cross-modal relationships [2]. Future research priorities include integrating pixel-level embeddings to address visual processing limitations [3], developing robust defense mechanisms against poisoning attacks [7], and creating standardized evaluation frameworks, as current benchmarks may not reflect real-world performance [9].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the latest things known about multimodal rag?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Multimodal Focus:** Does the study focus on multimodal RAG systems that process at least two different modalities (text, image, audio, video, etc.)?
- **RAG Architecture Focus:** Does the study have a specific focus on RAG (retrieval-augmented generation) architecture rather than general retrieval or generation approaches?
- **Substantial Investigation:** Does the study provide substantial investigation of multimodal RAG rather than only mentioning it in passing?
- **RAG Principles:** Does the study incorporate retrieval-augmented generation principles rather than focusing on traditional multimodal systems without RAG components?
- **Publication Completeness:** Is the study a complete publication with sufficient detail (not a conference abstract, workshop paper, or preliminary report lacking adequate depth)?
- **Technical Contribution or Empirical Evaluation:** Does the study present either: (1) novel architectures, methods, techniques, training methodologies, data fusion techniques, or optimization strategies for multimodal RAG, OR (2) empirical evaluation of multimodal RAG performance, including benchmarking, comparative analyses, evaluation metrics, datasets, or benchmarks?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Technical Approach:**
Extract the specific technical methods, architectures, or innovations for multimodal RAG described in this study, including:
 - Novel retrieval mechanisms or algorithms
 - Fusion strategies for combining different modalities
 - Architecture modifications or enhancements
 - Training or fine-tuning approaches specific to multimodal RAG
 - Integration methods for text, images, audio, video, or other modalities
 - Any algorithmic innovations that distinguish this approach from standard RAG or unimodal systems
- **Application Domain:**
Extract the specific application domain(s) or use case(s) where multimodal RAG is being applied or evaluated, such as:
 - Healthcare/medical diagnosis
 - Document understanding
 - Question answering
 - Industrial applications
 - Education
 - Scientific research
 - Security/safety applications Include details about the specific tasks within each domain (e.g., medical VQA, report generation, document analysis)

- **Modalities Integrated:**

Extract all types of data modalities that are integrated in the multimodal RAG system, including:

- Text (documents, reports, articles)
- Images (medical scans, charts, photographs)
- Audio/speech
- Video
- Tables/structured data
- Graphs/diagrams
- Other data types Note how these modalities are processed, retrieved, and combined in the RAG pipeline

- **Performance Results:**

Extract quantitative performance improvements or results achieved by the multimodal RAG approach, including:

- Accuracy improvements (with percentages and baselines)
- Comparison results vs. unimodal RAG or other baselines
- Success rates for specific tasks
- Error reduction metrics
- Factuality improvements
- Any other measurable performance gains Include specific numbers, confidence intervals, and what metrics were used

- **Key Challenges:**

Extract the main challenges, limitations, or problems identified with current multimodal RAG systems, such as:

- Cross-modal alignment issues
- Computational efficiency problems
- Hallucination or factual accuracy issues
- Scalability limitations
- Security vulnerabilities (e.g., poisoning attacks)
- Data quality or availability issues
- Integration complexity
- Evaluation difficulties Include both technical and practical challenges

- **Proposed Solutions:**

Extract specific solutions, techniques, or strategies proposed to address multimodal RAG challenges, including:

- Novel algorithms or methods to improve alignment
- Robustness enhancement techniques
- Efficiency optimization approaches
- Defense mechanisms against attacks
- Data augmentation or selection strategies
- Training improvements
- Architectural modifications Focus on concrete solutions rather than general recommendations

- **Evaluation Methods:**

Extract the evaluation approaches, datasets, benchmarks, and metrics used to assess multimodal RAG performance, including:

- Specific datasets used (names and characteristics)
- Evaluation benchmarks
- Metrics for measuring success (accuracy, F1, BLEU, etc.)
- Comparison baselines
- Human evaluation methods
- LLM-as-judge approaches
- Novel evaluation frameworks designed for multimodal RAG
- **Key Insights:**

Extract the main findings, insights, or conclusions about multimodal RAG from this study, including:

- What works well and what doesn't in multimodal RAG
- Surprising or counterintuitive results
- Best practices or design principles discovered
- Theoretical insights about multimodal retrieval and generation
- Practical lessons learned
- Future research directions identified
- Broader implications for the field Focus on novel knowledge or understanding gained about multimodal RAG specifically

Results

Characteristics of included studies

All 10 studies were published between 2024-2025, representing the most recent developments in multimodal RAG research. The studies encompass both primary research contributions and comprehensive surveys.

Study	Full text retrieved?	Study type	Primary focus	Application domain
Denys Yuvzhenko et al., 2025	Yes	Primary study [3]	ReAct agent-based multimodal RAG with multi-hop reasoning [3]	Healthcare/scientific research [3]
Peng Xia et al., 2024	Yes	Primary study [1]	Medical multimodal RAG (MMed-RAG) with domain-aware retrieval [1]	Medical diagnosis and treatment planning [1]
Monica Riedler et al., 2024	Yes	Primary study [6]	Integration of multimodal models into RAG for industrial applications [6]	Industrial domain (manufacturing, engineering, maintenance) [6]

Study	Full text retrieved?	Study type	Primary focus	Application domain
Yifei Shu et al., 2025	No	Survey [10]	RAG-based multimodal embedding techniques [10]	Medical image analysis and cross-modal content generation [10]
Mohammad Mahdi Abootorabi et al., 2025	Yes	Survey [8]	Comprehensive analysis of multimodal RAG systems [8]	Healthcare, software engineering, fashion, question answering, autonomous systems [8]
Sensen Gao et al., 2025	Yes	Survey [9]	Multimodal RAG for document understanding [9]	Finance, scientific research, social sciences, industrial document analysis [9]
Hyeonjeong Ha et al., 2025	Yes	Primary study [7]	Security vulnerabilities (MM-PoisonRAG attack framework) [7]	Question answering (VQA and chart understanding) [7]
Pei Liu et al., 2025	Yes	Primary study [5]	Hierarchical multi-agent multimodal RAG (HM-RAG) [5]	Scientific research and disaster response [5]
R. Rashmi et al., 2025	No	Primary study [4]	Modality-aware hybrid retrieval architecture (MAHA) [4]	Multimodal question answering and document understanding [4]
Zirui Guo et al., 2025	Yes	Primary study [2]	Unified RAG framework (RAG-Anything) [2]	Document understanding, financial analysis, academic research [2]

The studies span diverse application domains, with healthcare/medical applications receiving significant attention (4 studies) [1, 3, 8, 10], followed by document understanding (4 studies) [2, 4, 9], and industrial applications (1 study) [6]. Seven studies had full text available, while three were abstract-only.

Thematic analysis

Technical architectures and innovations

Recent multimodal RAG systems have evolved beyond simple text-based retrieval to incorporate sophisticated architectural designs. Several distinct architectural paradigms have emerged:

Agent-based architectures: Multiple studies demonstrate the effectiveness of agent-based approaches for multimodal RAG. Denys Yuvzhenko et al. implemented a ReAct agent logic system with multi-hop reasoning, using iterative reasoning processes with automatic synonymic paraphrasing [3]. Pei Liu et al. developed HM-RAG with a three-tiered architecture featuring specialized agents for decomposition, multi-source retrieval, and decision-making [5]. Agent-based multimodal RAG employs autonomous agents to manage retrieval and generation across modalities [9].

Graph-based approaches: Graph structures have emerged as a powerful method for organizing multimodal evidence. Sensen Gao et al. highlighted graph-based multimodal RAG as representing content as an explicit graph to enhance retrieval and reasoning [9]. R. Rashmi et al. proposed MAHA, which integrates dense vector retrieval with structured graph traversal using a modality-aware knowledge graph that encodes cross-modal semantics and relationships [4]. Zirui Guo et al. introduced dual-graph construction to capture both cross-modal relationships and textual semantics within a unified representation [2].

Hybrid retrieval mechanisms: Cross-modal hybrid retrieval combines structural knowledge navigation with semantic matching [2]. The integration includes planning-guided retrieval, agentic RAG, and feedback-driven iterative refinement [8]. Contrastive retrieval frameworks like MARVEL and Uni-IR improve inter-modal discrimination [8].

Domain-aware and adaptive systems: Peng Xia et al. introduced a domain-aware retrieval mechanism to handle different medical image domains more effectively, coupled with an adaptive method for selecting the optimal number of retrieved contexts using similarity scores [1]. This adaptive calibration approach dynamically adjusts retrieval based on relevance.

Modalities and integration strategies

Multimodal RAG systems integrate diverse data types, with varying approaches to processing and fusion:

Modality category	Specific types	Integration approaches	Studies
Text	Documents, reports, articles	Vector embeddings, semantic search	All studies
Visual	Medical scans, charts, photographs, diagrams	CLIP embeddings [6], textual descriptions [3], VLM-enhanced descriptions [5]	9 studies [1–9]
Structured data	Tables, equations	Separate vector collections [3], structured graph traversal [4]	6 studies [2–4, 9]
Graphs/diagrams	Charts, mathematical expressions	Textual descriptions [3], graph-based indexing [9]	5 studies [2–5, 9]
Audio/speech	Voice recordings	Multimodal encoders	2 studies [8]
Video	Motion sequences	Frame sampling	2 studies [8]

Systems employ two primary strategies for image integration: multimodal embeddings using models like CLIP [6] and generating textual summaries from images [6]. Monica Riedler et al. found that textual summaries from images present a more promising approach compared to multimodal embeddings, providing more opportunities for future advancements [6].

Performance achievements

Recent multimodal RAG systems demonstrate substantial improvements over text-only approaches:

Accuracy improvements: Denys Yuvzhenko et al. achieved 92% answer accuracy and 89% answer recall with 100% coverage across all modalities [3]. Peng Xia et al. reported an average improvement of 43.8% in factual accuracy for medical applications across five datasets [1]. Monica Riedler et al. observed that using both text and images increased Answer Correctness from approximately 60% to 80% [6].

Long-context performance: RAG-Anything demonstrated particularly strong performance on long documents, outperforming baselines by over 13 points for documents exceeding 100 pages [2]. On MMLongBench, accuracy gains ranged from 3.4 points for 11-50 pages to 9.3 points for 51-100 pages and 7.9 points for 101-200 pages [2].

Zero-shot capabilities: HM-RAG established state-of-the-art results in zero-shot settings on ScienceQA and Crisis-MMD benchmarks, improving answer accuracy by 12.95% and question classification accuracy by 3.56% over baseline RAG systems [5]. The framework exceeded human expert performance by 6.03% [5].

Medical domain achievements: MMed-RAG reduced the Copy-Reference rate to 28.19% and decreased the Over-Reliance rate from 43.31% to 8.38%, demonstrating improved alignment [1]. The system generalized across different backbone models, achieving 40.3% improvement over LLaVA-Med-1.0 [1].

Response efficiency: Average response times remained practical for interactive systems, with Denys Yuvzhenko et al. reporting approximately 5 seconds per query [3].

Challenges and limitations

Despite recent advances, multimodal RAG systems face significant challenges:

Cross-modal alignment issues: Current systems struggle with domain adaptation and exhibit modality biases, often relying too heavily on text [8]. Establishing coherent cross-modal correlations between visual and textual elements remains challenging [5]. Even pixel-level noise can disrupt alignment and lead to incorrect outputs [7]. Current RAG systems are fundamentally limited by their focus on unimodal textual data, neglecting multimodal information crucial in real-world documents [2].

Computational efficiency problems: High costs in video frame sampling and memory bottlenecks present significant barriers [8]. Graph-based methods face a trade-off between capturing high-level interactions and maintaining fine-grained information fidelity [5]. Processing large volumes of documents creates efficiency challenges [9]. Systems must balance recall with efficiency, as additional reasoning iterations increase latency with marginal recall improvements [3].

Security vulnerabilities: Hyeonjeong Ha et al. exposed critical vulnerabilities through knowledge poisoning attacks, demonstrating that Localized Poisoning Attack (LPA) achieved success rates up to 56% in manipulating generation, while Globalized Poisoning Attack (GPA) completely nullified generation with just a single irrelevant knowledge injection [7, 7]. The reliance on cross-modal representations makes multimodal RAG uniquely vulnerable to attacks that disrupt these representations [7].

Hallucination and factual accuracy: Current models suffer from generating non-factual responses, particularly in medical contexts [1]. Retrieving too much or too little information can lead to hallucinations [1]. LLMs lack domain-specific knowledge and are prone to hallucinations [6].

Image retrieval challenges: Image retrieval poses greater challenges than text retrieval [6]. Systems that rely on textual captions for images cannot answer questions about visual attributes like colors or shapes [3].

Explainability gaps: Systems struggle to attribute responses to specific sources within modalities, failing to pinpoint exact contributing elements across modalities [8].

Data quality and availability: There is a lack of high-quality labeled data for fine-tuning and distribution gaps between training and deployment data [1]. The absence of publicly available, domain-specific datasets affects reproducibility and generalizability [6]. Limited datasets for complex reasoning and adversarial examples hamper robust evaluation [8].

Scalability limitations: Building and traversing large multimodal graphs presents challenges [9]. Systems face retrieval speed-accuracy trade-offs and redundant computations in cross-modal fusion [8]. Single-source retrieval limits the ability to handle complex queries that require multiple data sources [5].

Solutions and innovations

Researchers have proposed numerous solutions to address these challenges:

Novel retrieval mechanisms: ChromaDB vector storage with OpenAI embedding generation models provides efficient retrieval [3]. Parallel multi-source retrieval across vector, graph, and web-based databases enables comprehensive coverage [5]. Maximum inner product search (MIPS) variants and adaptive quantization methods enhance efficiency [8]. Hybrid sparse-dense retrieval improves recall while maintaining speed [8].

Robustness enhancements: ReAct agent logic with explicit reasoning steps and iterative verification improves system robustness [3]. Hard negative mining and balanced sampling strategies during training enhance model resilience [8]. Hierarchical indexing and graph structures (VRAG-RL, MG-RAG) provide robust organization [9]. Consistency voting using ROUGE-L and BLEU metrics integrates answers from different modalities reliably [5].

Alignment improvements: RAG-based preference fine-tuning enhances cross-modality alignment and overall alignment with ground truth [1]. Contrastive retrieval frameworks like MARVEL and Uni-IR improve inter-modal discrimination [8]. Structure-aware evidence selection ensures relevant information is prioritized [9].

Efficiency optimizations: Indexing textual descriptions of images instead of processing pixel data reduces computational overhead [3]. Token merging and compressing page-level visual representations (Light-ColPali) improve efficiency [9]. Adaptive agent activation and lightweight proxy models reduce computational burden [9].

Data selection strategies: The Gap statistic method optimizes the number of retrieved contexts [1]. Augmentation techniques refine retrieved data for better interpretation [8].

Training improvements: Multistage processes capture cross-modal interactions effectively [8]. Image-to-caption and caption-to-image losses with sparsity regularization improve model learning [8].

Architectural innovations: Three-tiered architectures with specialized agents for decomposition, retrieval, and decision-making enable coordinated processing [5]. Dual-graph construction captures both cross-modal relationships and textual semantics [2]. Modality-aware query processing and cross-modal alignment systems ensure proper information flow [2].

Defense mechanisms: Expert model refinement resolves conflicting answers and ensures factual accuracy [5]. Privacy-preserving retrieval and verifiable generation protect against attacks [9].

Evaluation approaches

Multimodal RAG systems employ diverse evaluation methodologies:

Datasets and benchmarks: Studies use specialized datasets including the Global Tuberculosis Report 2024 with textual, graphical, and tabular data [3], medical datasets (MIMIC-CXR, IU-Xray, Harvard-FairVLMed, PMC-OA, Quilt-1M) [1], and document understanding benchmarks (DocBench, MMLongBench) [2]. The M²RAG benchmark provides a unified evaluation framework for both language quality and visual integration [8].

Performance metrics: Systems are evaluated using diverse metrics tailored to specific tasks:

- Retrieval metrics: Accuracy, recall, precision, F1-score, mean reciprocal rank (MRR), normalized discounted cumulative gain (nDCG) [8, 9]
- Text generation metrics: BLEU, ROUGE, METEOR, exact match [1, 8, 9]
- Visual quality: FID for image quality assessment [8]
- Medical domain: AUROC for VQA tasks [1]
- Multimodal-specific: Answer Correctness, Answer Relevancy, Text Faithfulness, Image Faithfulness, Text Context Relevancy, Image Context Relevancy [6]

LLM-as-judge approaches: Monica Riedler et al. developed a custom evaluation framework using LLMs to evaluate their own generated outputs across six specific metrics [6]. GPT-4o-mini serves as a judge for accuracy evaluations in document understanding tasks [2].

Comparison baselines: Studies compare against decoding-based methods (Greedy Decoding, Beam Search, DoLa, OPERA, VCD), multimodal RAG-based methods (MedDr, FactMM-RAG, RULE), and other open-source models [1]. State-of-the-art zero-shot LLMs, VLMs, and RAG-enhanced approaches serve as baselines [5].

Key insights and future directions

Research reveals several critical insights about multimodal RAG:

What works well: The three-tiered architecture with specialized agents for decomposition, retrieval, and decision-making represents a best practice in multimodal RAG design [5]. Combining textual, graph-relational, and web-derived evidence improves query understanding and accuracy [5]. Consistency voting and expert model refinement effectively integrate and refine answers [5]. Using modality-aware knowledge graphs and combining embeddings with explicit document structure proves highly effective [4].

Counterintuitive findings: Textual summaries from images outperform multimodal embeddings for integration [6], offering greater flexibility and potential for future advancements [6]. Using multiple images generally improves performance except for Image Faithfulness [6]. Graph construction is essential for performance, with chunk-only variants achieving significantly lower accuracy (60.0%) [2].

Design principles: Modality-aware reasoning is crucial for effective multimodal retrieval [4]. Integrating different modalities using structured approaches improves retrieval effectiveness [4]. Modular design and multi-source retrieval integration are key theoretical insights for improving systems [5]. Domain, retrieval modality, and granularity are critical factors in multimodal RAG [9].

Performance patterns: Performance gains increase with content length, becoming particularly pronounced on long-context documents where traditional approaches fail [2]. Finer-grained modeling within pages improves retrieval accuracy and robustness [9].

Architectural insights: Architectural fragmentation is a major issue in current systems, requiring specialized pipelines for each modality [2]. Dual-graph construction and cross-modal hybrid retrieval are effective strategies [2]. Current systems suffer from text-centric retrieval bias and rigid spatial processing patterns that need addressing [2].

Security implications: Multimodal RAG systems are vulnerable to knowledge poisoning attacks that can manipulate outputs by injecting misinformation [7]. Limited access to the retriever can significantly impact model performance, highlighting the importance of securing the retrieval process [7]. Robust defenses against knowledge poisoning attacks are urgently needed [7].

Future research priorities: Integrating pixel-level embeddings for images to address limitations in answering visual attribute questions [3]. Refining image retrieval and exploring fine-tuning-based approaches [6]. Developing layout-aware parsing mechanisms and integrating visual processing capabilities to improve robustness [2]. Incorporating reinforcement learning and human-aligned feedback to improve retrieval necessity assessment and modality selection [8]. Integrating data from diverse real-world sources to enhance situational awareness [8]. Addressing personalization mechanisms like user-specific retrieval, which remain underexplored [8].

Broader implications: Multimodal RAG enhances practical utility over traditional text-only solutions [3]. The technology improves reliability of medical LVLMs in clinical settings [1]. Systems can handle complex queries without prior training in zero-shot settings [5]. The field advances toward more robust and adaptable information retrieval and generation systems across diverse application domains [5]. Current benchmarks may not accurately reflect real-world performance, suggesting a need for better evaluation frameworks [9].

Synthesis

The research reveals substantial convergence on core principles while demonstrating domain-specific adaptations. All primary studies achieve significant performance improvements over text-only baselines, but the magnitude varies by application context and system design choices.

Architectural convergence: Three architectural paradigms dominate: agent-based systems (HM-RAG [5], ReAct-based [3]), graph-based approaches (MAHA [4], RAG-Anything [2]), and hybrid designs combining both. This convergence reflects a fundamental insight—multimodal content requires either autonomous coordination (agents) or explicit structural representation (graphs) to handle cross-modal dependencies effectively. The highest-performing systems (HM-RAG achieving 12.95% improvement [5], RAG-Anything with 13+ point gains on long documents [2]) integrate both paradigms, suggesting that pure agent-based or pure graph-based approaches have limitations.

Domain-context dependencies: Performance differences across domains reflect more than methodological variations—they reveal fundamental differences in multimodal integration requirements. Medical applications show the largest absolute gains (43.8% improvement in MMed-RAG [1]) because baseline text-only systems perform particularly poorly when visual evidence (scans, charts) is essential for diagnosis. Industrial applications show more modest improvements (60% to 80% accuracy increase [6]) because many queries can be answered adequately from text alone, with images providing supplementary rather than essential information. Document understanding shows performance gains that increase with document length (3.4 to 13+ points) [2], indicating that multimodal RAG's advantage grows when context windows exceed text-only model capabilities.

Image integration paradox: The counterintuitive finding that textual summaries of images outperform multimodal embeddings [6] initially appears to contradict the premise of multimodal RAG. However, this reflects current technical limitations rather than fundamental principles. Textual summaries work better because: (1) they leverage the superior performance of text embedding models versus multimodal encoders, (2) they avoid alignment errors between visual and text embeddings, and (3) they enable text-based reasoning that current LLMs excel at. Yet this approach fails for visual attribute questions (colors, shapes) [3], revealing its inherent limitations. The optimal solution likely involves hybrid approaches where textual summaries handle semantic content while direct image processing addresses visual-specific queries—a direction multiple studies recommend for future work [2, 3].

Security-performance trade-off: The vulnerability findings from MM-PoisonRAG (56% attack success rate [7], complete generation failure with single poisoned entry [7]) demonstrate that openness to external knowledge—multimodal RAG’s core strength—creates severe security risks. Systems achieving the highest performance (HM-RAG, RAG-Anything) use sophisticated multi-source retrieval that increases attack surface. The lack of robust defense mechanisms in current systems [7] suggests this trade-off remains unresolved. Systems must either sacrifice performance through defensive filtering (reducing recall) or accept vulnerability to poisoning attacks. This challenge is particularly acute for multimodal systems because cross-modal poisoning (injecting misleading text-image pairs) is more difficult to detect than text-only attacks.

Evaluation validity concerns: Studies use vastly different evaluation approaches—from domain-specific medical metrics [1] to LLM-as-judge frameworks [6] to traditional NLG metrics [9]. This heterogeneity makes cross-study comparisons unreliable. More critically, the concern that current benchmarks may not reflect real-world performance [9] casts doubt on reported improvements. Systems optimizing for benchmark performance may not handle actual deployment scenarios where documents have non-standard layouts, contain domain-specific terminology, or require understanding of subtle visual details. The field needs standardized, realistic evaluation frameworks before definitive conclusions about optimal architectures can be drawn.

Efficiency-complexity paradox: Complex mechanisms (graph traversal, multi-agent coordination, hybrid retrieval) offer marginal gains at increased computational cost [9]. Simple systems with textual summaries achieve 80% of the performance of sophisticated architectures [6] while using a fraction of the computational resources. Yet long-document performance [2] and zero-shot capabilities [5] improve substantially with architectural sophistication. This suggests a context-dependent optimization: simple approaches suffice for short documents with straightforward queries, while complex architectures become necessary for long-context, multi-hop reasoning tasks. Deployment decisions should be guided by expected query complexity rather than pursuing maximal performance across all scenarios.

References

1. Peng Xia, Peng Xia, Kangyu Zhu, et al (2024) MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models. International Conference on Learning Representations. <https://doi.org/10.48550/arXiv.2410.13085>
2. Zirui Guo, Xubin Ren, Lingrui Xu, et al (2025) RAG-Anything: All-in-One RAG Framework. arXiv.org. <https://doi.org/10.48550/arXiv.2510.12323>
3. Denys Yuvzhenko, Valentyn Chymshyr, V. Shymkovych, et al (2025) A Multimodal Retrieval-Augmented Generation System with ReAct Agent Logic for Multi-Hop Reasoning. Information, Computing and Intelligent systems. <https://doi.org/10.20535/2786-8729.6.2025.330777>
4. R. Rashmi, V. Upadhya (2025) Multimodal RAG for Unstructured Data:Leveraging Modality-Aware Knowledge Graphs with Hybrid Retrieval. arXiv.org. <https://doi.org/10.48550/arXiv.2510.14592>
5. Pei Liu, Xin Liu, Ruoyu Yao, et al (2025) HM-RAG: Hierarchical Multi-Agent Multimodal Retrieval Augmented Generation. Proceedings of the 33rd ACM International Conference on Multimedia. <https://doi.org/10.1145/3746027.3754761>

6. Monica Riedler, Stefan Langer (2024) Beyond Text: Optimizing RAG with Multimodal Inputs for Industrial Applications. arXivorg. <https://doi.org/10.48550/arXiv.2410.21943>
7. Hyeonjeong Ha, Qiusi Zhan, Jeonghwan Kim, et al (2025) MM-PoisonRAG: Disrupting Multimodal RAG with Local and Global Poisoning Attacks. arXivorg. <https://doi.org/10.48550/arXiv.2502.17832>
8. Mohammad Mahdi Abootorabi, Amirhosein Zobeiri, Mahdi Dehghani, et al (2025) Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-Augmented Generation. Annual Meeting of the Association for Computational Linguistics. <https://doi.org/10.48550/arXiv.2502.08826>
9. Sensen Gao, Shanshan Zhao, Xu Jiang, et al (2025) Scaling Beyond Context: A Survey of Multimodal Retrieval-Augmented Generation for Document Understanding. arXivorg. <https://doi.org/10.48550/arXiv.2510.15253>
10. Yifei Shu, Zejie Yu (2025) A Survey on RAG-Based Multimodal Embedding Techniques. International Conferences on Computers, Information Processing and Advanced Education. <https://doi.org/10.1109/CIPAE66821.2025.00070>